

Chapter 16: Abstract Vector Spaces

In Chapters 12 through 14, we worked algebraically with vectors in $\mathbb{R}^n$. Many of our results relied on the fact that vectors in $\mathbb{R}^n$ can be added and multiplied by scalars in a familiar way.

However, vectors in $\mathbb{R}^n$ is not the only set that has these properties. Other examples of sets that satisfy the same properties include: polynomials, differentiable functions, matrices, and others! Since so many useful sets exhibit the same properties, we will develop a name and theory surrounding these sets with addition and scalar multiplication that “play nice” with each other.

Definition: A (real) **vector space** consists of a nonempty set V of objects (called **vectors**) that can be added, that can be multiplied by a real number (called a **scalar** in this context), and for which certain axioms hold. If $\mathbf{v}$ and $\mathbf{w}$ are two vectors in V , their sum is expressed as $\mathbf{v} + \mathbf{w}$, and the scalar product of $\mathbf{v}$ by a real number a is denoted as $a\mathbf{v}$. The following axioms are assumed to hold for a vector space.

- A1. If $\mathbf{u}$ and $\mathbf{v}$ are any elements in V , then $\mathbf{u} + \mathbf{v}$ is in V . In this case, we say V is **closed** under vector addition.
- A2. $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$ for all $\mathbf{u}, \mathbf{v}$ in V .
- A3. $\mathbf{u} + (\mathbf{v} + \mathbf{w}) = (\mathbf{u} + \mathbf{v}) + \mathbf{w}$ for all $\mathbf{u}, \mathbf{v}, \mathbf{w}$ in V .
- A4. There exists an element $\mathbf{0}$ (written _____) in V such that $\mathbf{v} + \mathbf{0} = \mathbf{0} + \mathbf{v} = \mathbf{v}$ for any $\mathbf{v} \in V$. The vector $\mathbf{0}$ is called the **zero vector**.
- A5. For each $\mathbf{v}$ in V there exists an element $-\mathbf{v}$ in V such that $\mathbf{v} + (-\mathbf{v}) = -\mathbf{v} + \mathbf{v} = \mathbf{0}$. The vector $-\mathbf{v}$ is called the **negative** of $\mathbf{v}$ or **additive inverse** of $\mathbf{v}$.
- S1. If $\mathbf{v}$ is any element in V and a is any real number, then $a\mathbf{v}$ is in V . In this case, we say V is **closed** under scalar multiplication.
- S2. $a(\mathbf{u} + \mathbf{v}) = a\mathbf{u} + a\mathbf{v}$ for any $\mathbf{u}, \mathbf{v}$ in V and any a in $\mathbb{R}$.
- S3. $(a + b)\mathbf{v} = a\mathbf{v} + b\mathbf{v}$ for any $\mathbf{v}$ in V and any a, b in $\mathbb{R}$.
- S4. $a(b\mathbf{v}) = (ab)\mathbf{v}$ for any $\mathbf{v}$ in V and any a, b in $\mathbb{R}$.
- S5. $1\mathbf{v} = \mathbf{v}$ for any $\mathbf{v}$ in V .

The operation $\mathbf{v} + \mathbf{w}$ is called **vector addition** and, and the operation $a\mathbf{v}$ is called **scalar multiplication**.

Note: When using a “non-standard” vector addition and/or scalar multiplication, we will denote vector addition as $\oplus$ and scalar multiplication as $\odot$.

Example: Let V be the set of all positive real numbers, and define vector addition $\oplus$ by $u \oplus v = uv$ ($\oplus$ is ordinary multiplication) and define scalar multiplication $\odot$ by $c \odot v = v^c$. What is the zero vector $\mathbf{0}$ for V ?

(Note: you will show the rest of the vector space axioms are true for this $(V, \oplus, \odot)$ in discussion, so this is indeed a vector space!)

Note: We will often just use the phrase “vector space,” dropping the “real” and assuming our scalars are from the real numbers in this course. However, it is possible to have complex vector spaces, rational vector spaces, etc.!

Warning! From now on, “vector” will just mean an element of a vector space, not necessarily a directed line segment.

Example: The set $\mathbb{R}^n$ is a real vector space for any natural number n . This is the most common “day-to-day” vector space.

Example: The set of all $m \times n$ matrices with real entries with matrix addition as addition and scalar multiplication of a matrix by a real number as scalar multiplication is a vector space. We denote this vector space by M_{mn} , $M_{m \times n}$, or $\mathbb{R}^{m \times n}$. Prove that $M_{2 \times 2}$ is closed under addition and scalar multiplication. What is the zero vector in this vector space?

Example: A **polynomial** (in x) is a function expressible as

$$p(x) =$$

where $a_0, a_1, \dots, a_d$ are real numbers and d is a nonnegative integer. If $a_d \neq 0$, we say $p(x)$ has **degree** d .

Let $P(\mathbb{R})$ or P be the set of all polynomials (with real coefficients), and let $P_d(\mathbb{R})$ or P_d be the set of all polynomials (with real coefficients) of degree $\leq d$ together with the zero polynomial $p(x) = 0$, denoted $\mathbf{0}$.

Using the standard polynomial addition and scalar multiplication for polynomials, P is a vector space and each P_d is a vector space.

Let's walk through the proof of some of the axioms that P_2 (the set of polynomials of degree at most 2) is a vector space. We list them all here for completeness, but we'll choose a few to do in class.

(A1) If $\mathbf{u}$ and $\mathbf{v}$ are in V , then $\mathbf{u} + \mathbf{v}$ is in V .

(A2) $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$ for all $\mathbf{u}, \mathbf{v}$ in V .

(A3) $\mathbf{u} + (\mathbf{v} + \mathbf{w}) = (\mathbf{u} + \mathbf{v}) + \mathbf{w}$ for all $\mathbf{u}, \mathbf{v}, \mathbf{w}$ in V .

(A4) There exists an element $\mathbf{0}$ in V such that $\mathbf{u} + \mathbf{0} = \mathbf{0} + \mathbf{u} = \mathbf{u}$ for all $\mathbf{u}$ in V .

(A5) For each $\mathbf{u}$ in V there exists an element $-\mathbf{u}$ in V such that $\mathbf{u} + -\mathbf{u} = -\mathbf{u} + \mathbf{u} = \mathbf{0}$.

(S1) If $\mathbf{v}$ is in V and r is in $\mathbb{R}$, then $r\mathbf{v}$ is in V .

(S2) $r(\mathbf{u} + \mathbf{v}) = r\mathbf{u} + r\mathbf{v}$ for all $\mathbf{u}, \mathbf{v}$ in V and r in $\mathbb{R}$.

(S3) $(r + s)\mathbf{v} = r\mathbf{v} + s\mathbf{v}$ for all $\mathbf{v}$ in V and r, s in $\mathbb{R}$.

(S4) $r(s\mathbf{v}) = (rs)\mathbf{v}$ for all $\mathbf{v}$ in V and r, s in $\mathbb{R}$.

(S5) $1\mathbf{v} = \mathbf{v}$ for all $\mathbf{v}$ in V .

Example: Let V be the set of all ordered pairs of real numbers (x, y) with the operations

$$(x_1, y_1) \oplus (x_2, y_2) = (x_1, 0) \text{ and } c \odot (x, y) = (cx, cy).$$

Show that V is not a vector space.

Example: Let V be the set of all ordered pairs of real numbers (x, y) with the operations

$$(x_1, y_1) \oplus (x_2, y_2) = (x_1 + x_2, y_1 + y_2) \text{ and } c \odot (x, y) = (cx, y).$$

Show that V is not a vector space.

Example: Consider $V =$ non-negative real numbers, and define vector addition as $\mathbf{v} \oplus \mathbf{u} = |\mathbf{v} - \mathbf{u}|$, and scalar multiplication as $a \odot \mathbf{v} = |a|\mathbf{v}$. Does this form a vector space?

Below are some useful facts about the interaction of addition and scalar multiplication with the zero vector and the scalar zero.

Theorem: Let V be a vector space. For any $\mathbf{u}$ in V and a in $\mathbb{R}$, then

- (a) $0\mathbf{u} = \mathbf{0}$
- (b) $a\mathbf{0} = \mathbf{0}$
- (c) If $a\mathbf{u} = \mathbf{0}$, then $a = 0$ or $\mathbf{u} = \mathbf{0}$.
- (d) $(-1)\mathbf{u} = -\mathbf{u}$
- (e) $(-a)\mathbf{u} = -(a\mathbf{u}) = a(-\mathbf{u})$.

Proof of _____ (Pick one):

Chapter 17: Subspaces

Often a “big” vector space will have a (or many!) “little” vector space(s) inside of it.

Definition: Let V be a vector space and W be a nonempty subset of V (that is, $W \subseteq V$ with $W \neq \emptyset$). If W is a vector space with respect to the operations in V , then W is called a **subspace** of V .

Note: If $W \subseteq V$, then W automatically satisfies (A2), (A3), and (S2) – (S5), because

Furthermore, if W is closed under addition and scalar multiplication, then W satisfies (A4) and (A5), because

Therefore, if we know V is a vector space, we only need to check closure of W . More formally ...

Theorem: (Subspace Test) Let V be a vector space, and let W be a subset of V . Then W is a subspace of V if and only if the following conditions hold:

1. W is nonempty (i.e. W contains at least one vector)
2. If $\mathbf{u}$ and $\mathbf{v}$ are any vectors in W , then $\mathbf{u} + \mathbf{v}$ is in W .
3. If c is any real number and $\mathbf{u}$ is any vector in W , then $c\mathbf{u}$ is in W .

Example: The set $U = \left\{ \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} : x_1 x_2 = 0 \right\}$ in $\mathbb{R}^2$ is NOT a subspace of $\mathbb{R}^2$.

Example: Let W be the set of all vectors in $\mathbb{R}^3$ of the form $\begin{bmatrix} r \\ s \\ r+2s \end{bmatrix}$, where $r, s \in \mathbb{R}$.

Is W a subspace of $\mathbb{R}^3$?

Warning: To show a set IS a subspace, you NEED to use variables. To show a set is NOT a subspace, you should provide explicit (with numbers) counterexamples.

Example: Which of the following subsets of M_{22} with the usual operations of matrix addition and scalar multiplication are subspaces?

(a) W_1 is the set of all 2×2 matrices of the form

$$\begin{bmatrix} a & b \\ c & 0 \end{bmatrix}$$

(b) W_2 is the set of all 2×2 matrices of the form

$$\begin{bmatrix} a & b \\ c & 1 \end{bmatrix}$$

Example: Let W be the set of 2×2 symmetric matrices. Is W a subspace of M_{22} ?

Null Space and Image Space

Definition: Let A be an $m \times n$ matrix.

The **null space** of A , $\text{null}A$ is the set of vectors $\mathbf{x}$ in $\mathbb{R}^n$ which are solutions to $A\mathbf{x} = \mathbf{0}$, i.e.

$$\text{null}A = \{\mathbf{x} \in \mathbb{R}^n | A\mathbf{x} = \mathbf{0}\}.$$

This is also called the **solution space** of the homogeneous system $A\mathbf{x} = \mathbf{0}$.

The **image space** of A , $\text{im}A$, is the set of vectors $\mathbf{y}$ in $\mathbb{R}^m$ such that $A\mathbf{x} = \mathbf{y}$ has a solution, i.e.

$$\text{im}A = \{A\mathbf{x} | \mathbf{x} \in \mathbb{R}^n\}.$$

Theorem: For any $m \times n$ matrix A , the null space of A is a subspace of $\mathbb{R}^n$, and the image space of A is a subspace of $\mathbb{R}^m$.

Proof (we'll do one of the proofs in class):

Linear Combinations and Spanning Sets

Definition: Let $\mathbf{v}_1, \dots, \mathbf{v}_n$ be vectors in a vector space V . A vector $\mathbf{v}$ in V is a **linear combination** of $\mathbf{v}_1, \dots, \mathbf{v}_n$ if

$$\mathbf{v} =$$

for some $a_1, \dots, a_n \in \mathbb{R}$. The $a_1, \dots, a_n$ are called the **coefficients** of $\mathbf{v}_1, \dots, \mathbf{v}_n$. The set of all linear combinations of these vectors is called their **span**, and is denoted

$$\text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_n\} = \{a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + \dots + a_n\mathbf{v}_n \mid a_i \in \mathbb{R}\}$$

If every vector in V is a linear combination of the vectors in $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ (i.e. $V = \text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$), then these vectors are called a **spanning set** for V .

Example: Observe that every vector in $W = \left\{ \begin{bmatrix} r \\ s \\ r+2s \end{bmatrix} \mid r, s \in \mathbb{R} \right\}$ can be written as a linear combination of two (explicit) vectors:

Example: Consider $A = \begin{bmatrix} 5 & 1 \\ -1 & 1 \end{bmatrix}$ in M_{22} . Does A lie in $\text{span} \left\{ \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} -1 & 1 \\ 1 & 2 \end{bmatrix}, \begin{bmatrix} 0 & 2 \\ -2 & 1 \end{bmatrix} \right\}$?

Example: Let $\mathbf{v}_1 = x^2 + 2x + 1$ and $\mathbf{v}_2 = x^2 + 2$. Does $x^2 + 2x + 3$ lie in the span of $\{\mathbf{v}_1, \mathbf{v}_2\}$?

Example: $P_3 = \text{span}\{1, x, x^2, x^3\}$.

In fact, for any P_d , $P_d = \text{span}\{1, x, x^2, \dots, x^d\}$.

Example: Find a spanning set for $M_{2 \times 2}$.

Example: Find a spanning set for the null space of $A = \begin{bmatrix} 1 & -2 & -4 & 3 \\ 0 & -1 & -3 & 2 \\ 2 & 1 & 7 & -4 \end{bmatrix}$.

Example: Let $\mathbf{u}, \mathbf{v}$ be two vectors in a vector space V . Show that $\text{span}\{\mathbf{u}, \mathbf{v}\}$ is a subspace of V .

Theorem: Let $U = \text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ in a vector space V . Then:

1. U is a subspace of V containing each of $\mathbf{v}_1, \dots, \mathbf{v}_n$.
2. U is the “smallest” subspace containing these vectors in the sense that any subspace that contains each of $\mathbf{v}_1, \dots, \mathbf{v}_n$ must contain all of U .

Fact: Given vectors $\mathbf{v}_1, \dots, \mathbf{v}_k$ in a vector space V and a subspace U of V ,

$$\text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_k\} \subseteq U \text{ if and only if each } \mathbf{v}_i \text{ is an element of } U.$$

Theorem: Suppose S and S' are two sets of vectors in a vector space.

- a. If each vector in S can be expressed as a linear combination of the vectors in S' , then $\text{span} S$ is a subspace of $\text{span} S'$.
- b. If each vector in S can be expressed as a linear combination of the vectors in S' , *and* each vector in S' can be expressed as a linear combination of the vectors in S , then $\text{span} S$ and $\text{span} S'$ are the same subspace.

Example: Show that $P_2 = \text{span}\{2 - 2x^2, 4x + 2, 7x^2\}$.